动态随机一般均衡模型：理论与应用
第 8 章：小国开放经济

邓贵川

A 动态系统

A.1 非对称两国模型动态系统

- 弹性价格系统

$$\bar{C}_t^{\sigma}\bar{N}_t^{\varphi}=\mu A_t\bar{\hat{P}}_{Ht},\quad (\bar{C}_t^*)^{\sigma}(\bar{N}_t^*)^{\varphi}=\mu A_t^*\bar{\hat{P}}_{Ft}^*$$

$$\bar{Y}_t=(1-\alpha)\bar{\hat{P}}_{Ht}^{-\eta}\bar{C}_t+(1-s)o\Big(\frac{\bar{\hat{P}}_{Ht}}{\bar{Q}_t}\Big)^{-\eta}\bar{C}_t^*$$

$$\bar{Y}_t^*=(1-so)(\bar{\hat{P}}_{Ft}^*)^{-\eta}\bar{C}_t^*+so(\bar{\hat{P}}_{Ft}^*\bar{Q}_t)^{-\eta}\bar{C}_t$$

$$(\bar{C}_t^*)^{-\sigma}=\bar{C}_t^{-\sigma}\bar{Q}_t$$

$$(1-\alpha)\bar{\hat{P}}_{Ht}^{1-\eta}+\alpha\Big(\bar{\hat{P}}_{Ft}^*\hat{Q}_t\Big)^{1-\eta}=1$$

$$\alpha^*\Big(\frac{\bar{\hat{P}}_{Ht}}{\bar{Q}_t}\Big)^{1-\eta}+(1-\alpha^*)(\bar{\hat{P}}_{Ft}^*)^{1-\eta}=1$$

$$\bar{Y}_t=A_t\bar{N}_t,\quad \bar{Y}_t^*=A_t^*\bar{N}_t^*$$

- 家庭部门

$$\beta\mathbb{E}_t\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}}\frac{R_t}{\Pi_{t+1}}=1,\quad C_t^{\sigma}N_t^{\varphi}=\hat{W}_t$$

$$\beta\mathbb{E}_t\frac{(C_{t+1}^*)^{-\sigma}}{(C_t^*)^{-\sigma}}\frac{R_t^*}{\Pi_{t+1}^*}=1,\quad (C_t^*)^{\sigma}(N_t^*)^{\varphi}=\hat{W}_t^*$$

$$\left(\frac{C_t^{-\sigma}}{(C_t^*)^{-\sigma}}Q_t\right)^{\omega}\left[\frac{(1-\alpha)+\alpha(Q_t\hat{P}_{Ft}^*/\hat{P}_{Ht})^{1-\eta}}{(1-\alpha)+\alpha Q_t^{\eta}C_t^*/C_t}\right]^{1-\omega}=1$$

$$(1-\alpha)\hat{P}_{Ht}^{1-\eta}+\alpha\Big(\hat{P}_{Ft}^*Q_t\Big)^{1-\eta}=1$$

$$\alpha^*\Big(\frac{\hat{P}_{Ht}}{Q_t}\Big)^{1-\eta}+(1-\alpha^*)(\hat{P}_{Ft}^*)^{1-\eta}=1$$

- 厂商部门

$$Y_t=A_tN_t,\quad \frac{\hat{\dot{P}}_{Ht}}{\hat{P}_{Ht}}=\frac{1}{\mu}\frac{S_{1t}}{S_{2t}}$$

$$S_{1t}=(1-\alpha)\hat{P}_{Ht}^{-\eta}C_t^{1-\sigma}\frac{\hat{W}_t}{A_t}+\beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon}S_{1t+1}$$

$$S_{2t} = (1-\alpha)\hat{P}_{Ht}^{1-\eta}C_t^{1-\sigma} + \beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon-1}S_{2t+1}$$

$$\theta\Pi_{Ht}^{\epsilon-1} + (1-\theta)\left(\frac{\hat{\dot{P}}_{Ht}}{\hat{P}_{Ht}}\right)^{1-\epsilon} = 1$$

$$Y_t^* = A_t^* N_t^*, \quad \frac{\hat{\dot{P}}_{Ft}^*}{\hat{P}_{Ft}^*} = \frac{1}{\mu}\frac{S_{1t}^*}{S_{2t}^*}$$

$$S_{1t}^* = (1-\alpha^*)(\hat{P}_{Ft}^*)^{-\eta}(C_t^*)^{1-\sigma}\frac{\hat{W}_t^*}{A_t^*} + \beta\theta\mathbb{E}_t\left(\Pi_{Ft+1}^*\right)^{\epsilon}S_{1t+1}^*$$
$$S_{2t}^* = (1-\alpha^*)(\hat{P}_{Ft}^*)^{1-\eta}(C_t^*)^{1-\sigma} + \beta\theta\mathbb{E}_t\left(\Pi_{Ft+1}^*\right)^{\epsilon-1}S_{2t+1}^*$$

$$\theta(\Pi_{Ft}^*)^{\epsilon-1} + (1-\theta)\left(\frac{\hat{\dot{P}}_{Ft}^*}{\hat{P}_{Ft}^*}\right)^{1-\epsilon} = 1$$

- 均衡条件

$$Y_t = \left[(1-\alpha)\hat{P}_{Ht}^{-\eta}C_t + \alpha\left(\frac{\hat{P}_{Ht}}{Q_t}\right)^{-\eta}C_t^*\right]V_{Ht}$$
$$Y_t^* = \left[(1-so)(\hat{P}_{Ft}^*)^{-\eta}C_t^* + so\left(\hat{P}_{Ft}^*Q_t\right)^{-\eta}C_t\right]V_{Ft}^*$$

$$V_{Ht} = \theta\Pi_{Ht}^{\epsilon}V_{Ht-1} + (1-\theta)\left(\frac{\hat{\dot{P}}_{Ht}}{\hat{P}_{Ht}}\right)^{-\epsilon}$$
$$V_{Ft}^* = \theta(\Pi_{Ft}^*)^{\epsilon}V_{Ft-1}^* + (1-\theta)\left(\frac{\hat{\dot{P}}_{Ft}^*}{\hat{P}_{Ft}^*}\right)^{-\epsilon}$$

$$\frac{\hat{P}_{Ht}}{\hat{P}_{Ht-1}} = \frac{\Pi_{Ht}}{\Pi_t}, \quad \frac{\hat{P}_{Ft}^*}{\hat{P}_{Ft-1}^*} = \frac{\Pi_{Ft}^*}{\Pi_t^*}$$

- 货币当局

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R}\right)^{\rho_r}\left[\left(\frac{Y_t}{\bar{Y}_t}\right)^{\phi_y}\left(\frac{\Pi_{Ht}}{\Pi^0}\right)^{\phi_\pi}\right]^{1-\rho_r}$$
$$\frac{R_t^*}{R^*} = \left(\frac{R_{t-1}^*}{R^*}\right)^{\rho_r}\left[\left(\frac{Y_t^*}{\bar{Y}_t^*}\right)^{\phi_y}\left(\frac{\Pi_{Ft}^*}{\Pi^{*0}}\right)^{\phi_\pi}\right]^{1-\rho_r}$$

A.2 小国模型动态系统

- 弹性价格系统

$$\bar{Y}_t^* = \mu^{\frac{1}{\sigma+\varphi}}(A_t^*)^{\frac{1+\varphi}{\sigma+\varphi}}$$
$$(\bar{Y}_t^*)^{\sigma}\bar{Q}_t\bar{Y}_t^{\varphi} = \mu A_t^{1+\varphi}\bar{\hat{P}}_{Ht}$$
$$\bar{Y}_t = \left[(1-o)\bar{Q}_t^{\frac{1}{\sigma}} + o\bar{Q}_t^{\eta}\right]\bar{\hat{P}}_{Ht}^{-\eta}\bar{Y}_t^*$$
$$(1-o)\bar{\hat{P}}_{Ht}^{1-\eta} + o\bar{Q}_t^{1-\eta} = 1$$

- 大国动态系统

$$\beta\mathbb{E}_t\frac{(C_{t+1}^*)^{-\sigma}}{(C_t^*)^{-\sigma}}\frac{R_t^*}{\Pi_{t+1}^*} = 1, \quad (C_t^*)^{\sigma}(N_t^*)^{\varphi} = \hat{W}_t^*, \quad Y_t^* = A_t^*N_t^*$$
$$\hat{\dot{P}}_{Ft}^* = \frac{1}{\mu}\frac{S_{1t}^*}{S_{2t}^*}, \quad \theta(\Pi_t^*)^{\epsilon-1} + (1-\theta)(\hat{\dot{P}}_{Ft}^*)^{1-\epsilon} = 1$$

$$S_{1t}^* = (C_t^*)^{1-\sigma}\frac{\hat{W}_t^*}{A_t^*} + \beta\theta\mathbb{E}_t\left(\Pi_{t+1}^*\right)^{\epsilon} S_{1t+1}^*$$
$$S_{2t}^* = (C_t^*)^{1-\sigma} + \beta\theta\mathbb{E}_t\left(\Pi_{t+1}^*\right)^{\epsilon-1} S_{2t+1}^*$$
$$Y_t^* = C_t^* V_{Ft}^*, \quad V_{Ft}^* = \theta(\Pi_t^*)^{\epsilon} V_{Ft-1}^* + (1-\theta)(\hat{\hat{P}}_{Ft}^*)^{-\epsilon}$$
$$\frac{R_t^*}{R^*} = \left(\frac{R_{t-1}^*}{R^*}\right)^{\rho_r}\left[\left(\frac{Y_t^*}{\bar{Y}_t^*}\right)^{\phi_y}\left(\frac{\Pi_t^*}{\Pi^{*0}}\right)^{\phi_\pi}\right]^{1-\rho_r}$$

- 家庭部门

$$\beta\mathbb{E}_t\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}}\frac{R_t}{\Pi_{t+1}} = 1, \quad C_t^{\sigma}N_t^{\varphi} = \hat{W}_t, \quad (1-o)\hat{P}_{Ht}^{1-\eta} + oQ_t^{1-\eta} = 1$$

- 厂商部门

$$Y_t = A_t N_t, \quad \frac{\hat{\hat{P}}_{Ht}}{\hat{P}_{Ht}} = \frac{1}{\mu}\frac{S_{1t}}{S_{2t}}, \quad \theta\Pi_{Ht}^{\epsilon-1} + (1-\theta)\left(\frac{\hat{\hat{P}}_{Ht}}{\hat{P}_{Ht}}\right)^{1-\epsilon} = 1$$
$$S_{1t} = (1-o)\hat{P}_{Ht}^{-\eta}C_t^{1-\sigma}\frac{\hat{W}_t}{A_t} + \beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon}S_{1t+1}$$
$$S_{2t} = (1-o)\hat{P}_{Ht}^{1-\eta}C_t^{1-\sigma} + \beta\theta\mathbb{E}_t\Pi_{Ht+1}^{\epsilon-1}S_{2t+1}$$

- 均衡条件

$$Y_t = \left[(1-o)C_t + oQ_t^{\eta}C_t^*\right]\hat{P}_{Ht}^{-\eta}V_{Ht}$$

$$V_{Ht} = \theta\Pi_{Ht}^{\epsilon}V_{Ht-1} + (1-\theta)\left(\frac{\hat{\hat{P}}_{Ht}}{\hat{P}_{Ht}}\right)^{-\epsilon}, \quad \frac{\hat{P}_{Ht}}{\hat{P}_{Ht-1}} = \frac{\Pi_{Ht}}{\Pi_t}$$

$$\left[\frac{C_t^{-\sigma}}{(C_t^*)^{-\sigma}}Q_t\right]^{\omega}\left[\frac{(1-o)+o(Q_t/\hat{P}_{Ht})^{1-\eta}}{(1-o)+oQ_t^{\eta}C_t^*/C_t}\right]^{1-\omega} = 1$$

- 货币当局

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R}\right)^{\rho_r}\left[\left(\frac{Y_t}{\bar{Y}_t}\right)^{\phi_y}\left(\frac{\Pi_{Ht}}{\Pi^0}\right)^{\phi_\pi}\right]^{1-\rho_r}$$

B 求解模型稳态

B.1 稳态系统

B.1.1 非对称两国模型稳态系统

设稳态时 $A = A^* = 1$，$\Pi = \Pi^* = 1$，以及 $P_H = P_F = \xi P_H^* = \xi P_F^*$，因此，$Q = \hat{P}_H = \hat{P}_F^* = 1$，模型稳态系统为：

- 家庭部门

$$R = 1/\beta, \quad R^* = \frac{1}{\beta}, \quad C^{\sigma}N^{\varphi} = \hat{W}, \quad (C^*)^{\sigma}(N^*)^{\varphi} = \hat{W}^*$$

- 厂商部门

$$Y = AN, \quad Y^* = A^*N^*$$
$$\hat{\hat{P}}_H = \frac{1}{\mu}\frac{S_1}{S_2}, \quad \theta\Pi_H^{\epsilon-1} + (1-\theta)\hat{\hat{P}}_H^{1-\epsilon} = 1$$

$$S_1 = (1-\alpha)C^{1-\sigma}\frac{\hat{W}}{A} + \beta\theta\Pi_H^{\epsilon}S_1$$
$$S_2 = (1-\alpha)C^{1-\sigma} + \beta\theta\Pi_H^{\epsilon-1}S_2$$
$$\hat{\hat{P}}_F^* = \frac{1}{\mu}\frac{S_1^*}{S_2^*}, \quad \theta(\Pi_F^*)^{\epsilon-1} + (1-\theta)(\hat{\hat{P}}_F^*)^{1-\epsilon} = 1$$

$$S_1^* = (1-\alpha^*)(C^*)^{1-\sigma}\frac{\hat{W}^*}{A^*} + \beta\theta\left(\Pi_F^*\right)^{\epsilon}S_1^*$$
$$S_2^* = (1-\alpha^*)(C^*)^{1-\sigma} + \beta\theta\left(\Pi_F^*\right)^{\epsilon-1}S_2^*$$

- 均衡条件

$$Y = \left[(1-\alpha)C + (1-s)oC^*\right]V_H$$
$$Y^* = \left[(1-so)C^* + soC\right]V_F^*$$

$$V_H = \theta\Pi_H^{\epsilon}V_H + (1-\theta)\hat{\hat{P}}_H^{-\epsilon}$$
$$V_F^* = \theta(\Pi_F^*)^{\epsilon}V_F^* + (1-\theta)(\hat{\hat{P}}_F^*)^{-\epsilon}$$

$$\left(\frac{C^*}{C}\right)^{\sigma\omega} = \left[(1-\alpha) + (1-s)o\frac{C^*}{C}\right]^{1-\omega}$$
$$\Pi_H = \Pi, \quad \Pi_F^* = \Pi^*$$

- 潜在产出

$$\bar{C}^{\sigma}\bar{N}^{\varphi} = \mu A, \quad (\bar{C}^*)^{\sigma}(\bar{N}^*)^{\varphi} = \mu A^*$$
$$\bar{Y} = A\bar{N}, \quad \bar{Y}^* = A^*\bar{N}^*, \quad \bar{C}^* = \bar{C}$$
$$\bar{Y} = (1-\alpha)\bar{C} + (1-s)o\bar{C}^*$$
$$\bar{Y}^* = (1-so)\bar{C}^* + so\bar{C}$$

B.1.2 小国模型稳态系统

设稳态时 $A = A^* = 1$，$\Pi = \Pi^* = 1$，以及 $P_H = P_F = \xi P_H^* = \xi P_F^*$，因此，$Q = \hat{P}_H = \hat{P}_F^* = 1$, 以及 $V_H = V_F^* = 1$，模型稳态系统为：

$$Y^* = A^*N^*$$
$$\mu = \frac{S_1^*}{S_2^*}$$
$$S_1^* = \frac{(C^*)^{1-\sigma}}{1-\beta\theta}\frac{\hat{W}^*}{A^*}$$
$$S_2^* = \frac{(C^*)^{1-\sigma}}{1-\beta\theta}$$
$$R^* = R = 1/\beta$$
$$(C^*)^{\sigma}(N^*)^{\varphi} = \hat{W}^*$$
$$Y^* = C^*$$
$$C^{\sigma}N^{\varphi} = \hat{W}$$
$$Y = AN$$

$$\mu = \frac{S_1}{S_2}$$
$$S_1 = \frac{(1-o)C^{1-\sigma}}{1-\beta\theta}\frac{\hat{W}}{A}$$
$$S_2 = \frac{(1-o)C^{1-\sigma}}{1-\beta\theta}$$
$$Y = (1-o)C + oC^*$$
$$\Big(\frac{C^*}{C}\Big)^{\frac{\sigma\omega}{1-\omega}} = (1-o) + o\frac{C^*}{C}$$
$$\bar{Y} = \bar{Y}^* = \mu^{\frac{1}{\sigma+\varphi}} A^{\frac{1+\varphi}{\sigma+\varphi}} = \mu^{\frac{1}{\sigma+\varphi}} (A^*)^{\frac{1+\varphi}{\sigma+\varphi}}$$

B.2 稳态均衡

$$\Pi_H = \Pi_F^* = 1$$
$$R = R^* = 1/\beta$$
$$\hat{\dot{P}}_H = \hat{\dot{P}}_F^* = 1$$
$$\hat{W} = \hat{W}^* = \mu$$
$$V_H = V_F^* = 1$$
$$Y = C = Y^* = C^* = N = N^* = \mu^{\frac{1}{\sigma+\varphi}}$$
$$S_1 = \frac{1-\alpha}{1-\beta\theta}\mu^{\frac{1+\varphi}{\sigma+\varphi}}$$
$$S_2 = \frac{1-\alpha}{1-\beta\theta}\mu^{\frac{1-\sigma}{\sigma+\varphi}}$$
$$S_1^* = \frac{1-\alpha^*}{1-\beta\theta}\mu^{\frac{1+\varphi}{\sigma+\varphi}}$$
$$S_2^* = \frac{1-\alpha^*}{1-\beta\theta}\mu^{\frac{1-\sigma}{\sigma+\varphi}}$$

C 对数线性化

C.1 小国模型线性系统

$$\tilde{y}_t^* = \mathbb{E}_t\tilde{y}_{t+1}^* - \frac{1}{\sigma}\big(r_t^* - \mathbb{E}_t\pi_{t+1}^* - \bar{\hat{r}}_t^*\big)$$
$$\pi_t^* = \kappa(\sigma+\varphi)\tilde{y}_t^* + \beta\mathbb{E}_t\pi_{t+1}^*$$
$$r_t^* = \rho_r r_{t-1}^* + (1-\rho_r)\big(\phi_y\tilde{y}_t^* + \phi_\pi\pi_t^*\big)$$
$$c_t^* = y_t^* = \bar{y}_t^* + \tilde{y}_t^*$$
$$\bar{y}_t^* = \frac{1+\varphi}{\sigma+\varphi}a_t^*, \quad \bar{\hat{r}}_t^* = \frac{\sigma(1+\varphi)}{\sigma+\varphi}(\mathbb{E}_t a_{t+1}^* - a_t^*)$$
$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} + \mathbb{E}_t\bar{y}_{t+1} - \bar{y}_t - \Phi\big(r_t - \mathbb{E}_t\pi_{Ht+1}\big) + \frac{\sigma\eta(2-o)(\Psi-o)-\Psi}{\sigma\eta(2-o)+\Psi}(\mathbb{E}_t c_{t+1}^* - c_t^*)$$
$$\pi_{Ht} = \kappa\left\{\frac{1}{\Phi}(1+\varphi\Phi)(\tilde{y}_t+\bar{y}_t) - (1+\varphi)a_t + \frac{\sigma\eta(2-o)(\Psi-o)-\Psi}{\eta(2-o)(1-o+\Psi)}c_t^*\right\} + \beta\mathbb{E}_t\pi_{Ht+1}$$
$$r_t = \rho_r r_t + (1-\rho_r)\big(\phi_y\tilde{y}_t + \phi_\pi\pi_{Ht}\big)$$

$$\bar{y}_t = \frac{D(1+\varphi)}{\sigma + D\varphi} a_t + \frac{(1-D)\sigma}{\sigma + D\varphi} \bar{y}_t^*$$

$$q_t = \frac{1-o}{\eta o(2-o)} \frac{\Psi}{1-o+\Psi} y_t - \frac{1-o}{\eta o(2-o)} \frac{\Psi}{1-o+\Psi} c_t^*$$

其中，$\Psi = \frac{\eta o(2-o)[(\sigma+o)\omega - o]}{(1-2\eta+\eta o)o + (1-2o+2\eta o-\eta o^2)\omega}$，$\Phi = \frac{\eta(2-o)(1-o+\Psi)}{\sigma\eta(2-o)+\Psi}$。